

AI-Assisted Literature Review

From a research question → screening decisions → evidence

Today's Research Story

Imagine your group asks:

How does social media coverage influence fans' perceptions of athletes?

You search Google Scholar and find 500 articles.

500 candidate articles

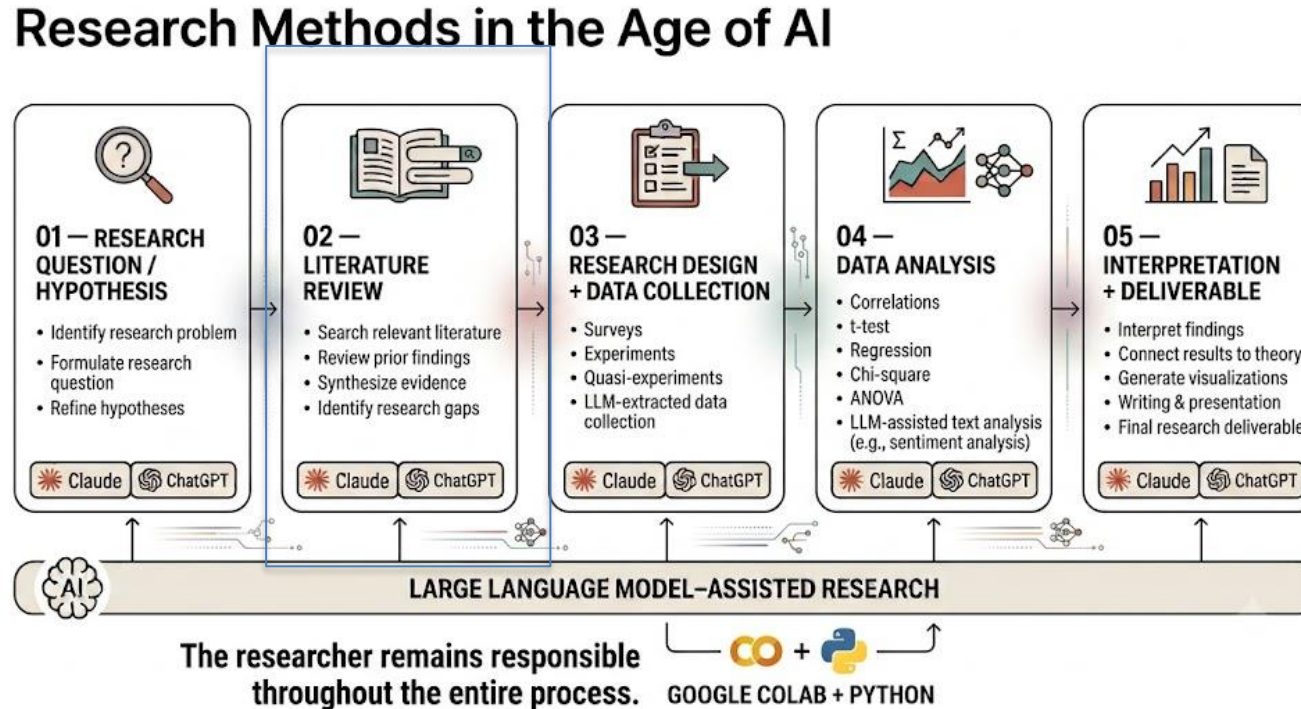


screen

25 articles to read closely

Today: How can Python + a language model help us do the FIRST PASS systematically?

Where Does This Fit in the Research Process?



Screening is a research decision: “Does this study help answer my question?”

First: What Is a Literature Review?

What is already known?

Find patterns across prior studies

What has been studied?

See populations, methods, variables, contexts

What is still missing?

Identify a gap your study can address

Literature review ≠ collecting papers. It is building an evidence-based argument.

The Screening Problem: Same Keywords ≠ Same Study

Research question: How does social media coverage influence fans' perceptions of athletes?

Clearly relevant

Examines athlete coverage on social media and measures fan perceptions.

Maybe relevant

Examines athletes' own posts and fan engagement, but not media coverage.

Not relevant

Examines social media advertising and sporting-event attendance.

AI needs YOUR definition of relevance.

Turn “Relevance” Into a Rule

Before using AI, write inclusion criteria.

- Population / context** → Sports fans or audiences
- Exposure / topic** → Social media coverage of athletes
- Outcome** → Perceptions, attitudes, trust, evaluation
- Study type** → Peer-reviewed empirical research

Research methods point: Operational definitions make judgments more consistent and reproducible.

Human Screening: What Would YOU Decide?

Abstract excerpt: “This survey examines how fans respond to athletes’ self-presentation on Instagram and whether perceived authenticity predicts identification with the athlete.”

**Using our criteria,
choose:**

INCLUDE

MAYBE

EXCLUDE

Then ask: Which criterion made you decide?

This is exactly the reasoning we will later ask a model to assist with.

Why Use Python + AI Instead of Only a Chatbot?

Chatbot

- ✓ Good for a few abstracts
- ✓ Manual copy/paste
- ✓ Harder to keep procedure identical

Python + model

- ✓ Repeat the same rule
- ✓ Process many abstracts
- ✓ Save every prediction

Research benefit: a documented, repeatable screening workflow.

Connect to Last Week: Python Already Knows the Pattern

1 STORE

abstracts in a list

2 REPEAT

the same task for
each abstract

3 DECIDE

apply a rule

4 SAVE

the result

```
for abstract in abstracts:  
    result = model(abstract)  
    save(result)
```

Read it in English: For each abstract → ask the model → save what it says.

How Can Python Talk to a Language Model?

1 Chat interface

You type into ChatGPT/Claude/Gemini.

2 API

Python sends text to a remote model.

3 Pretrained model in Python

Python loads an existing model and runs it in the workflow.

Today we practice #3 so you can see the workflow without managing an API.

What Is a Pretrained Model?

A model that has already learned patterns from data before we use it.

Before class

Someone trains the model on data.

In our notebook

We load the trained model.

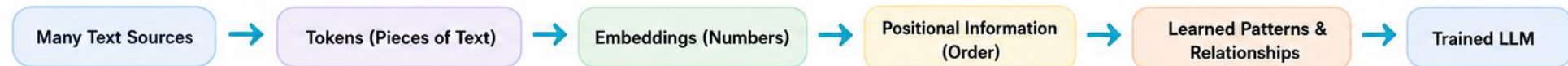
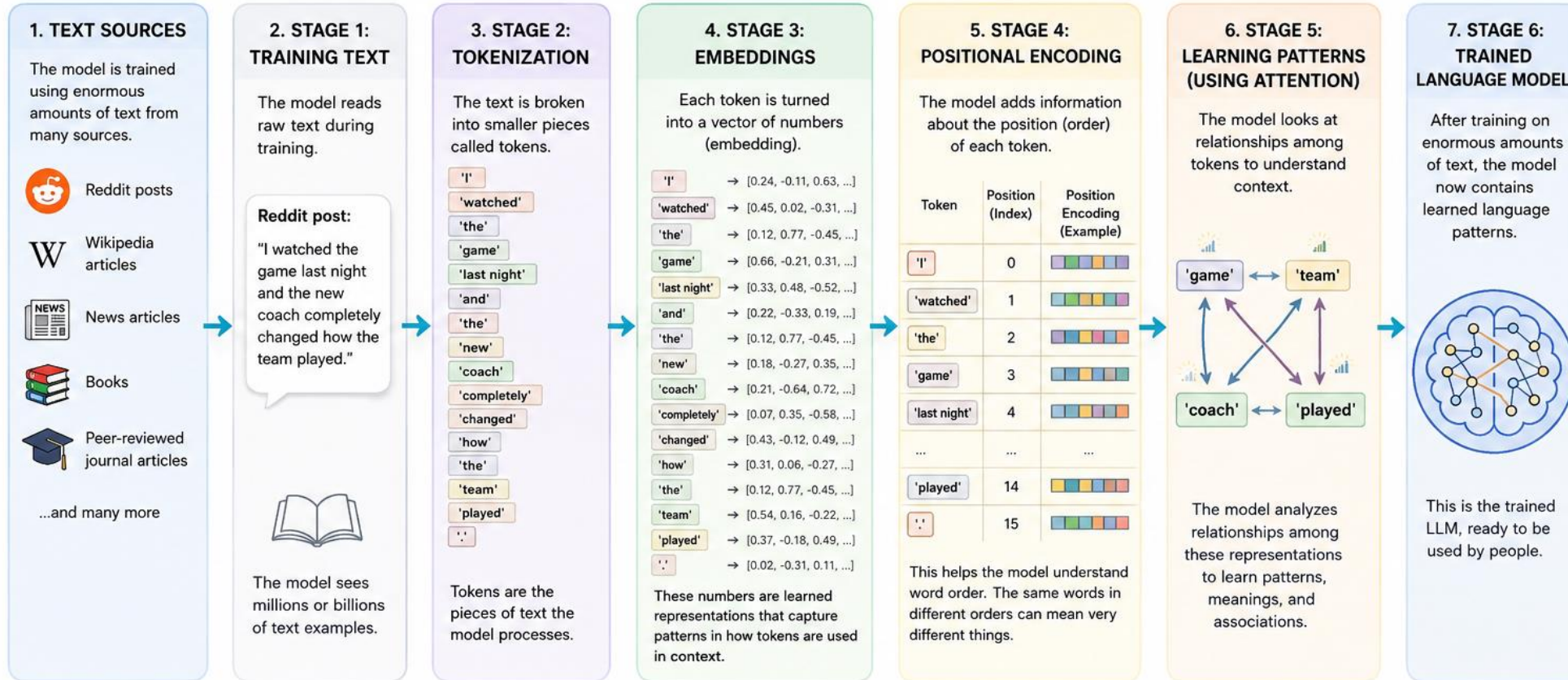
Our task

We give it new text and inspect its prediction.

Important: “pretrained” does NOT mean “appropriate for our research question.”

How a Large Language Model (LLM) Learns During Training

From raw text to a trained model that understands patterns in language



Training Time: The company trains the model using enormous amounts of text.

The result is a trained LLM that understands patterns in language and can be used later to respond to your prompts.



Key Takeaway

The model does not memorize these texts. It learns patterns and relationships.

Hugging Face = A Library of Models

Think of it like choosing a research instrument from a large catalog.

Model A

Sentiment: positive /
negative

Model B

Topic classification

Model C

Zero-shot classification

Question to ask first: Does the model's task match the construct I need?

A sentiment model cannot magically become a literature-relevance model.

Model Choice Is a Research Decision

Task

What prediction was this model built to make?

Labels

What answers can it return?

Training data

What kinds of text did it learn from?

Language

Does it support my documents?

Limits

How much text can it read?

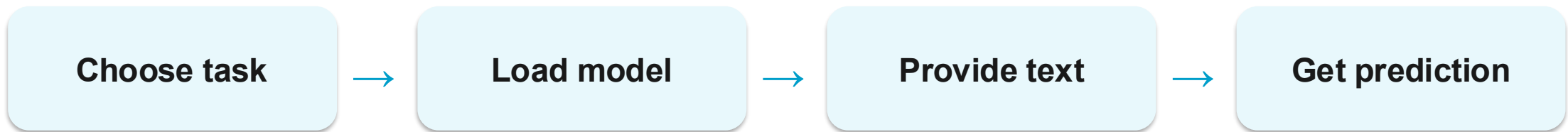
Performance

How was it evaluated?

Read the model card before treating the model like a measurement tool.

A Simple Starting Tool: pipeline()

pipeline() connects the pieces needed to use a pretrained model.



```
classifier = pipeline("sentiment-analysis")  
classifier("The coverage felt fair and informative.")
```

You do not need to memorize this. Read it as: “load a classifier → give it text.”

Easy Example First: Sentiment

Text: "The coverage felt fair and informative."



Prediction: POSITIVE score: 0.98

What does 0.98 mean?

The model is very confident in its own prediction.

It does NOT mean the prediction is 98% "true."

We still compare the prediction with human judgment.

Research Methods Warning: The Labels Define What Can Be Measured

Suppose the model only returns:

POSITIVE

NEGATIVE

Can it represent “neutral”?

No — not if neutral is not one of its labels.

Measurement lesson: A tool’s available categories constrain what it can tell you.

From One Text to Many Texts

```
abstracts = [  
    "Study A abstract...",  
    "Study B abstract...",  
    "Study C abstract..."  
]  
results = classifier(abstracts)
```

Abstract A



Prediction A

Abstract B



Prediction B

Abstract C

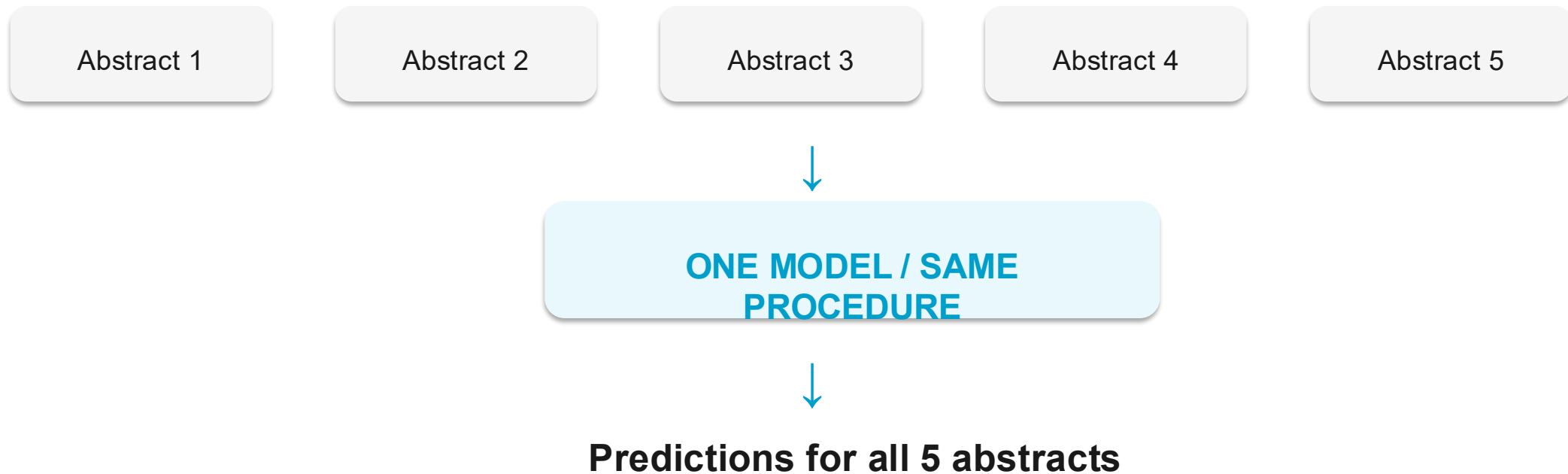


Prediction C

Same model + same procedure + many inputs = systematic repetition.

This Is Called Batching

Batch = multiple inputs processed together.



But Our Real Goal Is Relevance — Not Sentiment

Sentiment was only a simple example to learn the workflow.

Sentiment task

positive / negative

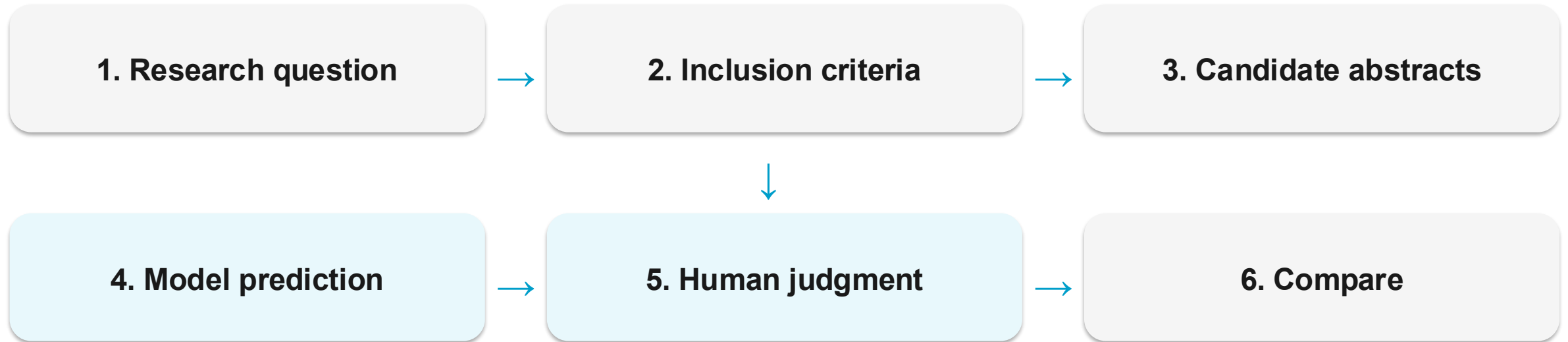
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Our screening task

relevant / maybe / not relevant

Next class: we will build a model-assisted triage workflow around YOUR research question and abstracts.

What Will the Screening Workflow Look Like?



**The comparison is the research-methods part: When does AI agree with humans?
When does it fail?**

Example: Human vs. Model Decisions

Abstract	Model	Human	What do we learn?
A: media coverage + fan trust	INCLUDE	INCLUDE	Agreement
B: athlete posts + engagement	INCLUDE	MAYBE	Possible over-inclusion
C: sports ads + attendance	EXCLUDE	EXCLUDE	Agreement

A useful AI tool is not one we simply trust — it is one we evaluate.

How Do We Evaluate the Model?

1

Agreement

How often do model and human make the same decision?

2

False inclusion

Model says include; human says exclude.

3

False exclusion

Model says exclude; human says include.

4

Patterns of error

Which kinds of abstracts confuse the model?

For screening, false exclusions can be especially costly: you may throw away relevant evidence.

A Model Can Also Miss Part of an Abstract

Models have input limits.

[Background] ... [Methods] ... [Results] ... [Conclusion: directly relevant to fan trust]

Model reads only this portion



important ending may be truncated

Research implication: check model documentation and inspect what text is actually being processed.

The Researcher Remains in the Loop

YOU define

the research question and inclusion criteria

MODEL assists

with repetitive first-pass classification

YOU verify

ambiguous cases and errors

YOU report

how the tool was used and evaluated

Automation changes the workflow. It does not remove methodological responsibility.

Quick Check: Which Is the Better Research Practice?

A

Pick the first model that runs.

B

Choose a model whose task and labels fit the screening goal.

C

Accept a 0.99 score as proof the answer is correct.

D

Compare model decisions with human decisions.

Best answers: B and D. Why?

What I Want You to Remember Today

1

Screening is a research decision, not a keyword-matching exercise.

2

Python helps repeat and save the same procedure across many texts.

3

A pretrained model is a tool; model choice affects what you can measure.

4

A prediction score is not truth. Human validation still matters.

Next Class: AI-Assisted Literature Screening Lab

Bring your group CSV of abstracts.

1. Load your abstracts
2. Apply a consistent screening rule
3. Get model-assisted predictions
4. Make human judgments
5. Compare model vs. human

Goal: use AI as a research assistant — then evaluate the assistant.